rag -- 2026-11-05

5 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P1] RAGCompass: Generalizable Retriever Routing via Reranking -- RAGCompass provides a clear, quantitative demonstration of a generalisable routing mechanism that directly improves answer quality and offers extensive analysis of utility-based supervision, making it the most informative single paper for understanding retrieval quality, grounding, and system design in RAG.

Themes

Several papers address how to improve the retrieval component of RAG systems beyond a static, one-size-fits-all retriever. RAGCompass introduces a utility-driven router that directly evaluates each retriever's passage set and selects the best one per query, achieving a 28% relative F1 gain over the strongest fixed retriever and showing strong zero-shot generalisation to new corpora and retrievers [P1]. In a similar spirit, EAGR proposes an evidence-guided, graph-free expansion that adaptively grows a semantic frontier based on LLM-assessed evidence, boosting multi-hop QA accuracy by 17% over vanilla RAG and by 4 points over the best graph-aware baseline [P2]. Both approaches share the idea of dynamic, query-dependent retrieval, but they differ in implementation: RAGCompass relies on contrastive reranking of whole passage sets, while EAGR uses metadata-driven neighborhood lookup without materialising a graph.

Graph structure is another axis of comparison. Weaver builds a dual-layer graph that incrementally incorporates user-preference evidence, yielding large gains on personalised RAG benchmarks (up to 16-point ACC improvements) by explicitly modelling intent dependencies and memory-conditioned retrieval trajectories [P3]. ArcRAG, by contrast, investigates whether a single retriever can remain effective across radically different graph constructions, showing that a construction-agnostic retriever works for both schema-guided and open-IE graphs, though absolute performance still varies with the graph choice [P5]. These works highlight a tension between tightly coupled graph-specific retrievers and more portable retrieval interfaces.

Faithfulness and grounding are examined in RAG-TRACE, which quantifies how often retrieved ECG criteria lack numeric verification and demonstrates that deterministic verification can prune unsupported statements while only modestly reducing overall diagnostic F1 (from 0.804 to 0.795) [P4]. This paper shifts focus from pure retrieval quality to downstream factual correctness, underscoring the need for evaluation benchmarks that capture grounding failures. Across the set, evaluation gaps remain: several studies (e.g., ArcRAG) omit detailed quantitative results, and others (e.g., Weaver) lack runtime cost analysis, making it hard to compare efficiency trade-offs alongside accuracy.

5 theme sentence(s) checked against the paper summaries.

At a glance

[P1] RAGCompass: Generalizable Retriever Routing via Reranking · relevance 0.50
RAGCompass routes queries to the most suitable retriever by directly evaluating the retrieved passages, yielding a 28% relative F1 gain over the best fixed retriever and showing strong generalization to unseen corpora, retrievers, and readers.

[P2] EAGR: Evidence-Guided Adaptive Graph Retrieval · relevance 0.39
EAGR introduces an evidence-guided, graph-free retrieval approach that adaptively expands a semantic frontier and achieves the highest LLM-based multi-hop QA accuracy across three benchmarks, surpassing vanilla RAG by 17 % and the strongest graph-aware baseline by 4 percentage points.

[P3] Weaver: Weaving Preference Evidence through Incremental Reasoning for Personalized Retrieval-Augmented Generation · relevance 0.38
Weaver introduces a dual-layer graph that incrementally retrieves and composes user-preference evidence, achieving state-of-the-art accuracy on personalized RAG benchmarks.

[P4] RAG-TRACE: Tracing Measurement Evidence and Diagnostic Completeness in AI-Generated ECG Reports · relevance 0.38
Retrieval-augmented ECG report generation often adds unsupported diagnostic criteria, but deterministic measurement verification can remove most of these without greatly harming diagnostic performance.

[P5] ArcRAG: Agentic Compositional Retrieval over Knowledge Graphs · relevance 0.37
ArcRAG shows that a construction-agnostic retriever can remain effective across two very different graph constructions for multi-hop QA, though absolute accuracy still depends on the graph used.

[P1] RAGCompass: Generalizable Retriever Routing via Reranking

Valentin Six, Mufei Li, Dongqi Fu, Yinglong Xia et al. · arXiv 2611.05596v1 · https://arxiv.org/abs/2611.05596v1 · relevance 0.50

TL;DR: RAGCompass routes queries to the most suitable retriever by directly evaluating the retrieved passages, yielding a 28% relative F1 gain over the best fixed retriever and showing strong generalization to unseen corpora, retrievers, and readers.

Problem
Different retrieval-augmented generation (RAG) systems rely on multiple retrievers, yet no single retriever excels on all queries, creating a need for a router that can select the optimal retriever for each query [1]. Existing routers use only query-side features and a fixed set of retrievers, which prevents them from handling unseen retrievers or new corpora [1][2].

Method
RAGCompass reads the actual passage sets returned by each retriever, making them directly comparable regardless of their source [1]. It contrastively fine-tunes a reranker against the downstream answer utility (F1) produced by each evidence set, which provides the bulk of the performance gain [1]; a small comparison head is added on top of the reranker to capture additional signal [1][4]. The approach contains no retriever-specific parameters and relies on utility supervision rather than relevance supervision [4].

Key results
- Answer F1 improves by 28% relative to the best fixed retriever [1].
- Utility supervision outperforms relevance supervision by 2.11 percentage points [4].
- Adding the comparison head raises performance by about 1.4 pp over a scalar scorer alone (0.3552 ± 0.0026 vs. 0.3419) [7].
- Per-dataset gains over the best fixed retriever are: HotpotQA +12.40 pp, 2Wiki +12.21 pp, MuSiQue +4.78 pp, NQ +0.42 pp; these capture 25.0 %, 21.8 %, 10.2 %, and 1.2 % of the headroom to the oracle respectively [8].
- Pooled across datasets, the routed system gains +7.70 pp over the global best fixed retriever (SPLADE-v3, F1 0.2782) [8].
- Inference overhead for the routing decision alone is 3.9 % (0.222 s/query) compared with 5.687 s for a full retrieve-and-answer pipeline [8][5].
- The experiments required roughly 700 GPU-hours (~475 h for labeling utility, ~200 h for training and evaluation) on NVIDIA L40S GPUs [8].

Limitations
Utility is defined using a single reader (Llama-3.1-8B-Instruct) and a fixed retrieval depth of five passages; only one alternative reader (Qwen2.5-7B-Instruct) is tested, so evidence quality cannot be disentangled from reader fit [5]. Deeper retrieval depths are untested because they would require re-running all retrievers and regenerating labels [5]. All results use one backbone size (Qwen3-Reranker-0.6B), leaving scalability to larger models unmeasured [5]. The three query-only baselines are re-implementations that

may differ from the originals, and the MoR baseline is a simplified variant [5]. Reported inference overhead accounts only for the routing decision, not the cost of running all retrievers, which--though parallelizable--adds additional compute [5][8].

Sources

[1] arXiv 2611.05596v1.pdf, arXiv abstract

[2] arXiv 2611.05596v1.pdf, page 3

[4] arXiv 2611.05596v1.pdf, page 9

[5] arXiv 2611.05596v1.pdf, page 10

[7] arXiv 2611.05596v1.pdf, page 14

[8] arXiv 2611.05596v1.pdf, page 20

[P2] EAGR: Evidence-Guided Adaptive Graph Retrieval

Sonam Gupta, Vignesh P, Vishwajeet Kumar, Adnan Qidwai et al. · arXiv 2611.01910v1 · https://arxiv.org/abs/2611.01910v1 · relevance 0.39

TL;DR: EAGR introduces an evidence-guided, graph-free retrieval approach that adaptively expands a semantic frontier and achieves the highest LLM-based multi-hop QA accuracy across three benchmarks, surpassing vanilla RAG by 17 % and the strongest graph-aware baseline by 4 percentage points.

Problem
Multi-hop reasoning with Retrieval-Augmented Generation requires the retrieval process to adapt as intermediate evidence reveals new information needs, but existing methods rely on explicit graph structures or iterative query reformulation, incurring graph-construction or query-time overhead[1].

Method
EAGR builds an implicit semantic graph from compact passage metadata stored in a standard VectorDB, where each passage is paired with a set of semantic descriptors (entities and key concepts) and two passages are linked if they share a descriptor[2]. The system starts from dense-retrieval seed passages, then an LLM iteratively assesses the accumulated evidence, selects promising descriptor sets, expands the frontier via metadata-based neighborhood lookup, and stops when sufficient evidence is gathered, all without materializing a graph or rewriting the query[1][5]. Implementation uses NV-EMBED-V2 for dense retrieval, descriptors extracted with LLAMA-3.3-70B-INSTRUCT, retrieval prompts with QWEN3-8B, an initial seed of k = 4 passages and a budget of B = 30 passages[4].

Key results
Across HotpotQA, 2WikiMultiHopQA, and MuSiQue, EAGR attains the highest average LLM-Accuracy of 67.1 %, which is 3.9 points above the strongest baseline (LinearRAG = 63.2) and 7.3 points above vanilla RAG[4]. It improves over vanilla RAG by 17 % and over the strongest graph-aware baseline by 4 percentage points on average[1]. Per-benchmark LLM-Accuracy scores are 77.4 (HotpotQA), 78.1 (2WikiMultiHopQA), and 45.8 (MuSiQue)[4], with corresponding String Accuracy scores of 67.4, 80.3, and 38.4 respectively[4]. An ablation study shows that removing edge selection drops Str-Acc to 40.9 on HotpotQA, while removing edge reranking reduces LLM-Acc to 66.3 on average, confirming the contribution of each component[4]. Retrieval gains are reported as largest on 2Wiki, where EAGR reaches a Recall@5 of 95.5 % (exact figure for other datasets not provided)[4].

Limitations
All EAGR components are used off-the-shelf without task-specific fine-tuning, and the authors note that future work could explore component-specific training and stronger foundation models to improve retrieval quality and practical viability[6].

Sources

[1] arXiv 2611.01910v1.pdf, arXiv abstract

[2] arXiv 2611.01910v1.pdf, page 3

[4] arXiv 2611.01910v1.pdf, page 7

[5] arXiv 2611.01910v1.pdf, page 8

[6] arXiv 2611.01910v1.pdf, page 9

[P3] Weaver: Weaving Preference Evidence through Incremental Reasoning for Personalized Retrieval-Augmented Generation

Linyao Zheng, Zhifang Mao, Xuhang Shi · arXiv 2611.05900v1 · https://arxiv.org/abs/2611.05900v1 · relevance 0.38

TL;DR: Weaver introduces a dual-layer graph that incrementally retrieves and composes user-preference evidence, achieving state-of-the-art accuracy on personalized RAG benchmarks.

Problem
Personalized retrieval-augmented generation must extract user preferences from long, fragmented interaction histories, where evidence is often missing or outdated, leading to unresolved or misapplied preferences [1][2].

Method
Weaver builds a Dual-Layer Retrieval and Reasoning Graph comprising an upper Intent Dependency Graph for tracking information gaps and a lower Memory-Conditioned Retrieval Trajectory Graph that records how retrieved fragments guide subsequent queries [1][3]. The system performs Memory-Conditioned Incremental Retrieval guided by the dependency graph and applies Scope-Aware Compositional Reasoning to interpret evidence within its temporal and situational context [1][5].

Key results
- On the PersonaMem benchmark, Weaver attains 86.42 % accuracy (ACC), 78.57 % F1, and 73.84 % exact match (EM), surpassing the strongest baseline (MemCoE) by 16.01, 10.51, and 10.53 percentage points respectively [5].
- On PrefEval, Weaver reaches 91.10 % ACC, 85.37 % F1, and 82.61 % EM, improving over the best baseline (MemoryBank) by 7.56, 3.21, and 3.53 points respectively [5].
- Ablation studies show that removing the dependency graph drops ACC by 13.97 points on PersonaMem and 14.60 points on PrefEval; removing incremental retrieval reduces ACC by 7.14 and 7.21 points; removing compositional reasoning lowers ACC to 84.27 (-2.15) on PersonaMem and 84.16 (-6.94) on PrefEval [5].
- When provided identical candidate memories, Weaver's compositional reasoning yields ACC gains of 2.50 points (single memory), 5.00 points (two memories), and 8.33 points (three or more memories) over a direct reader [6].

Limitations
The paper does not report the computational cost or runtime of Weaver despite posing RQ3 about cost, so those metrics are unavailable [4]. It also evaluates only on four benchmarks (PersonaMem, PrefEval, MuSiQue, WebWalker) and does not provide results on broader domains or user studies [4].

Sources

[1] arXiv 2611.05900v1.pdf, arXiv abstract

[2] arXiv 2611.05900v1.pdf, page 2

[3] arXiv 2611.05900v1.pdf, page 4

[4] arXiv 2611.05900v1.pdf, page 5

[5] arXiv 2611.05900v1.pdf, page 6

[6] arXiv 2611.05900v1.pdf, page 7

[P4] RAG-TRACE: Tracing Measurement Evidence and Diagnostic Completeness in AI-Generated ECG Reports

Elyas Irankhah, Shahed Salehzehi, Parisa Hosseini · arXiv 2611.01409v1 · https://arxiv.org/abs/2611.01409v1 · relevance 0.38

TL;DR: Retrieval-augmented ECG report generation often adds unsupported diagnostic criteria, but deterministic measurement verification can remove most of these without greatly harming diagnostic performance.

Problem
Retrieval-augmented generation (RAG) can insert ECG criteria that lack patient-specific measurement evidence, raising concerns about the reliability of AI-generated reports [1]. The study seeks to quantify how often retrieved criteria are unsupported and to assess whether verification can mitigate this limitation [6].

Method
The authors analyzed 1,083 human-validated PTB-XL patient records using a library of 47 criteria cards, of which 33 lacked numeric verification rules [1][2]. Language models (e.g., GPT-4.1-mini, Qwen3-8B) generated structured ECG reports both with and without retrieval; three retrieval strategies (keyword overlap, TF-IDF, BM25) each returned the top-three ranked cards [2]. Criterion applicability was checked against Marquette 12SL measurements, with Uni-G serving as an independent reference; a deterministic verification rule withdrew any criterion without a numeric check or that failed the 12SL predicate [1][4].

Key results
Keyword retrieval produced untestable or 12SL-disconfirmed criteria in 46.8 % of GPT-4.1-mini reports [4]. Each report contained on average 0.891 applied criteria: 0.590 lacked numeric checks, 0.007 failed 12SL predicates, and 0.294 were supported [4]. Deterministic withdrawal removed 647 criterion applications (639 without numeric checks, 8 with failed numeric predicates) while preserving 268 of 269 Uni-G-supported applications for GPT-4.1-mini and 274 of 282 for Qwen3-8B [4]. Restricting retrieval to 14 testable criteria lowered the unsupported-criterion rate to 13.8 % [1]. Under matched instructions, retrieval increased the proportion of reports with unsupported criteria from 17.4 % to 31.1 % but left hypertrophy recall unchanged at 62.6 % [1]. Micro-F1 dropped modestly from 0.804 to 0.795 in a different-schema comparison [6].

Limitations
The evaluation relied solely on published PTB-XL diagnostic labels without independent clinical review of the generated reports [6]. The predefined affirmation rule likely undercounted diagnostic assertions, and the exploratory label-aware analysis excluded 138 sentences with judge disagreement [6]. Model comparisons were constrained by incomplete JSON generation for Llama, a small sample of 112 DeepSeek cases, and the use of only 14 numeric-only criteria in some arms [6].

Sources

[1] arXiv 2611.01409v1.pdf, arXiv abstract

[2] arXiv 2611.01409v1.pdf, page 3

[4] arXiv 2611.01409v1.pdf, page 8

[6] arXiv 2611.01409v1.pdf, page 14

[P5] ArcRAG: Agentic Compositional Retrieval over Knowledge Graphs

Jyotin Goel, Abhinav Arun, Tejas Prakash Agarwal, Bhaskarjit Sarmah et al. · arXiv 2611.06481v1 · https://arxiv.org/abs/2611.06481v1 · relevance 0.37

TL;DR: ArcRAG shows that a construction-agnostic retriever can remain effective across two very different graph constructions for multi-hop QA, though absolute accuracy still depends on the graph used.

Problem
Graph-based retrieval-augmented generation (GraphRAG) typically couples its retrieval strategy to the specific knowledge-graph construction pipeline, and it is unclear whether a single retriever can work well when the underlying graph changes [1].

Method
ArcRAG induces a query-local graph interface at inference time--extracting node labels, relation descriptors, and property keys reachable from the entities mentioned in a question--and then plans and executes multi-hop traversals as Cypher queries [1]. The system keeps the retriever's configuration fixed while building each of MuSiQue, HotpotQA, and TriviaQA as (i) a schema-guided graph with typed nodes and a closed relation set, and (ii) an open-information-extraction graph with untyped nodes and free-text relations [1]. It is positioned as "induced" along the portability axis, contrasting with prior systems that rely on fixed or curated schemas [2].

Key results
The authors report that retrieval need not be co-designed with the graph: the query-time interface induction keeps a construction-agnostic retriever effective when the construction changes, although the choice of construction still influences absolute accuracy [1, 4]. The excerpts do not provide concrete numeric scores, accuracy gains, or detailed comparisons with baselines.

Limitations
The study tests only two maximally different constructions and does not establish transfer to curated ontologies or graphs with different node granularities [6]. Because the constructions differ in both structure and provenance, the observed accuracy gap conflates these factors and is not disentangled [6]. The paper reports a single run per configuration, uses temperature-1.0 sampling for the answer model, and does not quantify run-to-run variance [6]. Comparisons with prior GraphRAG systems rely on reported figures rather than shared experimental setups, and the benchmarks are encyclopedic QA, so reliability for regulated domains remains unverified [6]. The authors also note that graph construction still affects accuracy, so they do not claim that retrieval and construction never benefit from co-design [6].

Sources

[1] arXiv 2611.06481v1.pdf, arXiv abstract

[2] arXiv 2611.06481v1.pdf, page 3

[4] arXiv 2611.06481v1.pdf, page 7

[6] arXiv 2611.06481v1.pdf, page 9